“To fly! To live as airmen live! Like them to ride the skyways from horizon to horizon, across rivers and forests... Ah! That is life.”

Henri Mignet
L’Aviation de l’Amateur, 1934

SMALL IS BEAUTIFUL

SOME OF THE MOST SPECTACULAR FLYING FEATS OF MODERN TIMES HAVE BEEN ACHIEVED IN LIGHTWEIGHT AIRCRAFT OR EVEN BALLOONS

On **December 14, 1986,** at 8:00am, former fighter pilot Dick Rutan and his partner Jeana Yeager set out from Edwards Air Force Base in California to fly nonstop around the world, without in-flight refueling. Their propeller-driven aircraft, *Voyager*, had been designed by Dick Rutan’s younger brother, aeronautical engineer Burt Rutan. The pilots were to make the journey of over 26,000 miles (42,000km) at an average speed of only 110mph (175kph), in a cockpit and cabin that were barely 2ft (0.6m) wide.

Voyager was made of low-density high-strength carbon composites, so light that the airframe weighed only 939lb (426kg). That weight was virtually doubled by the two engines, but dwarfed by the fuel load, over 7,000lb (3,200kg) at takeoff. The wing was 111ft (34m) long – longer than that of a Boeing 727 – but only 3ft (1m) across, a shape offering maximum lift and minimum drag. Like the fuselages, it was packed with fuel.

Dick Rutan has said that before the flight he was not looking forward to “being inside this flailing carbon structure, a long way from home, listening to engines and hoping they keep running.” *Voyager* had never taken off with a full fuel load before the around-the-world attempt, and only just made it, using almost all of Edwards Air Force Base’s 15,000ft (4,500m) runway, with the tips of the fuel-filled wings dragging on the ground. It was very fragile in its fueled-up state and could not have landed if it had gotten into trouble. Rutan and Yeager benefited from excellent state-of-the-art instruments, including an autopilot, weather radar, and the latest navigational aids, but once they headed out over the Pacific they were on their own, at grips with the unknown. Weather proved a terrifying hazard as they dodged thunderheads and a typhoon. Over Africa on the fifth day, they had to fly up to 20,000ft (6,000m) to go above a line of thunderstorms, and both suffered from hypoxia through not breathing enough of their oxygen supply.

PEGASUS XL SE ULTRALIGHT
Ultralights are often in effect powered hang gliders, controlled by pivoting the wing, although some have an airplane-style control system. Flying one of these machines creates an experience that is in some ways similar to that of the early flight pioneers.